

# **THE IMPACT OF ICT REGULATIONS ON THE ACHIEVEMENT GAP IN MATHEMATICS: A CROSS-SECTIONAL ANALYSIS**

**Econometrics, Spring 26**

**Group 5**

Sara Lacerda, 62649

Yelyzaveta Lukianova, 56397

José Nunes, 56871

Henrique Sommer, 56334

## Abstract

More than just leading to better grades, does stricter school-level ICT regulation help reduce the performance differential between poorer and richer students? This topic becomes remarkably important in an age where the claim that children spend more time in front of screens than in nature or socializing does not seem to be so absurd anymore. At the same time, it opens doors to growing inequalities, as stable and more technologically educated households tend to better control their children's digital exposure, while families from poorer backgrounds may end up neglecting online threats, such as the algorithm trap, or the addictive properties of broader ICT devices. Drawing on PISA 2022 cross-sectional data from ten developed economies and nearly 82,000 observations, we developed four Weighted-Least-Squares models to assess whether ICT regulation can work as an effective tool to mitigate these counterproductive inequality effects and how is this relationship moderated by demographic, institutional, behavioural and country fixed effects. One of our main findings indicates that there is a statistically significant positive, yet relatively small, association between ICT Regulation and the reduction in the achievement gap in Mathematics test scores. Furthermore, although lower-income students do not happen to suffer more in terms of grade-penalty than higher-income ones from ICT induced distractions, their social and economical background remains one of the most powerful predictors of academic performance, even after accounting for factors like school type or individual ICT usage patterns. Lastly, we emphasized the need for policies targeting undisciplined and rule-free digital exposure, while acknowledging that their effect on non-median students may be more unpredictable.

## Table of Contents

|       |                                            |    |
|-------|--------------------------------------------|----|
| 1     | Introduction .....                         | 4  |
| 2     | Literature Review .....                    | 5  |
| 3     | Econometrics Models and Data.....          | 7  |
| 3.1   | Dependent Variable .....                   | 7  |
| 3.2   | Interest Variables .....                   | 8  |
| 3.3   | Control Variables.....                     | 9  |
| 3.3.1 | Student demographics.....                  | 9  |
| 3.3.2 | Personal ICT habits .....                  | 9  |
| 3.3.3 | Institutional characteristics .....        | 10 |
| 3.3.4 | Country fixed effects .....                | 11 |
| 3.4   | Model Variations .....                     | 11 |
| 4     | Violations of Assumptions .....            | 12 |
| 4.1   | Linearity in Parameters (MLR.1) .....      | 12 |
| 4.2   | Random Sampling (MLR.2).....               | 12 |
| 4.3   | No Perfect Collinearity (MLR.3).....       | 12 |
| 4.4   | Zero Conditional Mean (MLR.4) .....        | 12 |
| 4.5   | Homoskedasticity (MLR.5).....              | 13 |
| 4.6   | Normality in Error Term (MLR.6) .....      | 13 |
| 5     | Results .....                              | 14 |
| 5.1   | Main Regression and Model Comparison ..... | 14 |
| 5.2   | The Interest Variables.....                | 15 |
| 5.3   | Residual Analysis .....                    | 16 |
| 6     | Conclusion.....                            | 17 |
| 7     | Bibliography.....                          | 18 |
| 8     | Appendix .....                             | 19 |

# 1 Introduction

Over the last few decades, societies all around the world experienced massive shifts induced by technological developments that transformed everyday habits, social interactions, and the way people spend their time. Due to the widespread ownership of laptops and smartphones, screen time now constitutes a significant part of our daily lives: According to a Statista report, in 2024, the daily time spent using the internet by online users worldwide averaged six hours and 35 minutes.

At the same time, the “digital divide”, at least in western economies, has become less about discrepancies in the access to these devices and more focused on the change in utilization patterns between socioeconomic groups. In a research paper released in 2008, the pioneer in digital inequality, Eszter Hargittai, suggested that while socioeconomically disadvantaged students tend to use ICT for recreational purposes, which displace study time, their more affluent peers leveraged its usage to enhance career opportunities and steepen their learning curve.

In a moment where countries like France and the United Kingdom are debating smartphone bans, the goal of this project is to study whether school-level ICT regulations are an effective tool for social equity by protecting students who lack digital literacy and stricter control at home. To that end, we conducted a cross-sectional analysis of ten developed countries in the West and Asia, built on nearly 82,000 observations from the PISA 2022 database, and used Mathematics scores as a proxy for academic success. Our choice for Mathematics scores as the dependent variable is motivated by the 2022 PISA report, which recorded the largest decline in Mathematics scores in the history of the assessment — an unfortunate ongoing trend — and the fact that disadvantaged students were found to be seven times less likely to achieve basic mathematical proficiency, underscoring the urgency of policies that could narrow this gap.

Furthermore, our approach consisted of developing four Weighted Least Squares models, each adding another layer of complexity to the previous model and serving a specific purpose: transitioning from a baseline Direct Impact of ICT variables on scores to a Moderating Effect of socioeconomic and ICT-related interaction variables; followed by a Demographic and Institutional Filter to mitigate potential Omitted Variable Bias; and culminating in Country Fixed Effects, our most precise and final regression.

## 2 Literature Review

The analysis of the existing literature on digital technology and academic performance guided the construction of our models. We reviewed studies that supported the choice of our main variables of interest, particularly school-level ICT regulation and its interaction with socioeconomic status.

Hargittai (2008), in her seminal work on digital inequality, argued that the same levels of ICT exposure produce different outcomes depending on socioeconomic background. While disadvantaged students tend to use digital devices for recreational purposes that displace study time, their higher-income peers leverage technology to enhance learning and career opportunities. This divergence in usage patterns, rather than access alone, compounds existing academic inequalities and constitutes the core theoretical motivation for our research question.

Ji and Pang (2023) investigated the causal effect of social media use on student learning outcomes using PISA 2018 data across 50 countries. Their results reveal significant cross-country heterogeneity, with negative effect observed in the United States and United Kingdom but positive associations in Mexico and Turkey, highlighting that the national context plays a decisive role in shaping the relationship between digital use and performance. Recognising this, we introduced country fixed effects in our final specification to avoid drawing misleading conclusions from aggregate estimates.

Maghnouj et al. (2025), in an OECD policy paper drawing on PISA 2022 data, found that excessive digital use displaces time otherwise available for homework and that disadvantaged students face greater difficulty regulating their digital leisure outside school. These findings motivated the inclusion of behavioural controls for ICT Leisure and ICT distraction, ensuring that any estimated effect of regulation is not confounded by students' pre-existing digital habits at home or in class.

Bohnert and Gracia (2024), in their paper "Digital Distraction, Attention Regulation, and Inequality", formalised the Differential Vulnerability hypothesis, proposing that lower socioeconomic students get distracted more easily and experience a steeper performance decline for any given level of distraction. This mechanism directly motivated the construction of our interaction term between ICT Distraction and low socioeconomic status, as well as the need to control separately for low socioeconomic status, rather than relying solely on the continuous ESCS index.

Lubienski and Lubienski (2006) found that once student demographics are controlled for, the performance advantage of private schools disappears in most cases,

motivating our including of *private\_school* as an institutional control to prevent school type from confounding the estimated effect of ICT regulation.

Sirin (2005) showed that socioeconomic status is strongly related to academic achievement, especially in Mathematics. This is particularly important for our study, as it supports the inclusion of a student's socioeconomic background as one of the main control variables in our models.

Taken together, the literature establishes that recreational ICT use is negatively associated with Mathematics performance, and that socioeconomic status, as well as demographic, behavioral, institutional, and country-level factors must all be accounted for to isolate the equity effect of digital policies on the achievement gap.

### 3 Econometrics Models and Data

The data used to create this report came from the PISA 2022 database, which includes answers from students, teachers, parents and schools to questionnaires sent to a range of schools' representative of the different educational systems and economic realities across the OECD. Its focus is to measure 15-year-old students' proficiency in Mathematics, Reading and Science, but it also covers their opinions about learning, home environment and school life, among other dimensions.

As PISA assessments are conducted triennially and the latest report comes from 2022, we had to rely on this data to ensure that our analysis is as recent and relevant as possible. Although our initial objective was a cross-sectional study of all OECD countries present on the database, we ended up with a sample of only ten countries and 81,850 observations due to lack of data in some countries — we wanted to ensure that all the observations were complete in terms of the relevant variables to use in our models, even if it can increase the risk of sample selection bias.

After carefully considering the factors and nuances that the literature review considered important and acknowledging the limitations of our database when it comes to the dimensions we intended to measure, we built a model that explains how much school-level ICT rules help lower income students reduce the achievement gap in Mathematics scores compared to their more affluent peers. In that final model, which results from a rational buildup of three previous models, we included controls for aspects such as student behaviour, socioeconomic conditions, school type and country fixed effects, in attempt to minimize the likelihood of Omitted Variable Bias.

#### 3.1 Dependent Variable

We chose Mathematics scores ( $test\_sco$ ) as our dependent variable across all our models. The items on PISA's surveys from which these scores result assign 25% of score points to mathematical reasoning and 75% to mathematical problem-solving, divided into three processes: Formulating Situations Mathematically, Employing Mathematical Concepts and Procedures, and Interpreting and Evaluating Mathematical Outcomes.

Items' weights depend on complexity, and scores are measured on a continuous scale anchored to 2003 normalized average ( $\mu_{test\_sco} = 500$ ). Ever since, scores can be interpreted by the number of standard deviations ( $\sigma_{test\_sco} = 100$ ) they are from the 2003 mean:

$$test\_sco = (Z * 100) + 500$$

These results can be translated into six proficiency levels, where the Baseline (Level 2), the critical threshold for active societal participation, corresponds to the 420-482 range; the Proficient (Level 4) to the 545-607 range; and the Elite (Level 6) to scores above 669. Moreover, on this scale, an increase of 20 points is equivalent to another full year of schooling.

As Table 2 (Appendix) indicates, our sample average is 492.2, exactly 19.8 points above the OECD average of 472.4. We believe this difference is attributed to the fact that our sample includes some of the OECD's richest countries and none of its poorest (such as Colombia or Costa Rica), which is expected because high-income countries usually provide more data than lower-income ones and wealth is a key predictor of academic performance.

### 3.2 Interest Variables

The interaction between school-level ICT regulations and low socioeconomic status (*inter\_reg\_lowses*) is our primary variable of interest. One of the variables of this interaction (*ICT\_Regulation*) measures, on a 1-6 scale, the regulatory intensity at the school level on the usage of technological devices, integration of these devices for teaching and learning purposes, and prevention of ICT-related distraction sources during classroom hours. Higher values reflect stricter digital control. Furthermore, as only school principals provide this data, it does not vary between students at the same school.

The other variable (*low\_ses*) is a dummy that results from selecting the bottom 25% of students according to the PISA index of economic, social, and cultural status (*ESCS*). This index is standardized so that its mean is zero and standard deviation is one (in our sample, by observing Table 2, we have that  $\bar{x}_{ESCS} \approx 0.04$  and  $s_{ESCS} \approx 0.93$ ). So, *low\_ses* takes value one when the low *ESCS* condition is respected and zero otherwise. However, it is worth noting that after weighting the observations for the sake of representativeness, our *low\_ses* sample mean dropped to 0.22, suggesting that the proportion of disadvantaged students was initially overrepresented.

Rather than assuming a linear relationship between regulation and grades across the different socioeconomic groups, our working hypothesis implies the isolation of regulations' effect on disadvantaged students' Mathematics scores. That is why *low\_ses* is a necessary construction (and *ESCS* would not be enough by itself) to observe a potential reduction in the achievement gap promoted by stricter rules.



### 3.3 Control Variables

Besides the variables previously presented (*ESCS*, *low\_ses*, *ICT\_Regulation* and the interaction *inter\_reg\_lowses*), our models include several other control variables of four different natures to capture factors that might be correlated with both Mathematics scores and ICT regulations.

#### 3.3.1 Student demographics

Within this category, we have a dummy variable for gender (*female*). Despite recording higher school performance globally, a report conducted by the OECD based on PISA 2022 data shows that boys outperformed girls in Mathematics by nine score points on average. Furthermore, according to a study promoted by the European Commission, boys are more frequently top performers in Mathematics than girls in all EU countries. Therefore, by controlling for gender, we eliminate the possibility of confounding effects driven by different gender composition across schools.

In Table 2, the sample mean of roughly 0.50 for *female* (and Kurtosis of -2.00) indicates that the sample is balanced in terms of gender distribution.

#### 3.3.2 Personal ICT habits

One of our controls, *ICT\_Distraction*, measures, on a 1-6 scale, how often students get distracted by digital devices during lessons. As the OECD report clearly highlights, digital distraction is one of the primary drivers of the recent Mathematics scores decline. If we had decided not to include this control, our models would not distinguish between an increase in regulation and a decrease in distraction, taking the risk of overestimating regulatory effectiveness. Not to mention that our primary predictor, *inter\_reg\_lowses*, would likely capture the effect that disadvantaged students are more easily distracted, as suggested by Bohnert and Gracia (2024).

*ICT\_Distraction* median of three demonstrates that digital distraction happens with some frequency to the typical student in our sample, while the 0.38 skewness signals there is a small group who are extremely distracted, even though its distribution is nearly symmetrical.

On the other hand, *ICT\_Leisure* accounts for the frequency of using digital devices for fun outside the classroom. The scale works in the same way as with the previous variable — it varies between one and six and higher values reflect more usage — and it prevents, for instance, ICT regulations from mistakenly capturing the effect of

students who dedicate more time to study at home because their highly regulated school environment is associated with less technology use for leisure.

For this variable, a sample median of four and (slightly) negative skewness of -0.45 suggest that not only is using ICT for leisure quite frequent, but also that there is a high concentration of heavy ICT users. Lastly, its platykurtic distribution shows that there is not a single “uniform” type of student, but rather a wide variety of digital lifestyles.

Our last behavioural control is the interaction between digital distraction and low socioeconomic status, called *inter\_dist\_lowses*. Based on the *Differential Vulnerability* hypothesis suggested by Bohnert and Gracia (2024), disadvantaged students may experience a steeper grade decline for the same level of distraction and, on top of that, get distracted more easily. Hence, by accounting for the fact that digital distraction hurts this group more, we can isolate the policy effect from the pre-existing differences in terms of student discipline and reaction to digital exposure.

Descriptive statistics for this interaction show a very high positive skewness of 2.59, which makes sense as it is only different from zero for a small portion of our sample (recall that  $\bar{x}_{low\_ses} = 0.22$ ), and a kurtosis of 5.75 with a heavy tail representing the subset of students who are from a poorer background and get extremely distracted.

### 3.3.3 Institutional characteristics

Expectedly, controlling for school type through a dummy variable (*private\_school*), where zero means that the school is Public and one that the school is Private, is essential to account for the difference in rules and resources they often exhibit. It also addresses factors such as parental motivation regarding education and potential school self-selection bias.

Literature suggests that after considering socioeconomic status, the impact of school type on Mathematics grades is statistically insignificant (Lubienski & Lubienski (2006)). Additionally, it is important to highlight that the PISA 2022 report shows no clear difference in performance trends between educational systems that experienced longer or shorter closure periods during COVID-19.

In our sample, roughly 18% of students attended private schools ( $\bar{x}_{private\_school} \approx 0.18$ ), which we believe provides enough variation to control for school type specific effects.

### 3.3.4 Country fixed effects

Our last set of controls includes a dummy variable for each country, except for Germany, that is activated (for example,  $Country\_USA = 1$ ) when the observation is recorded in that country. It is of the greatest importance to ensure that our results and conclusions are not mistakenly driven by different educational systems, national curricula and cultures.

The choice to leave Germany without an assigned variable was not an accident, but a necessary condition to avoid perfect multicollinearity between our countries' regressors.

## 3.4 Model Variations

To observe the way our primary estimators respond to changes in model specifications and how consistent our results are across different control stages, we adopted a four-model approach:

Model 1 focuses on the direct relationship between Mathematics scores, ICT-related individual variables, both in terms of regulation and student behaviour, and the socioeconomic status continuous index. It establishes our baseline case.

Model 2 follows by introducing the dummy for disadvantaged students, the interaction variable of interest and the interaction between disadvantaged students and digital distraction to test for a possible distraction penalty for this specific group.

Model 3 works as a filter for demographic and institutional characteristics. It is useful to assess if our moderating effect is significant after controlling for possible gender or school type gaps.

Model 4 finally addresses unobserved national factors and constitutes our most precise and final regression. Hence, it is the one that we will analyse most carefully, and whose extensive equation form is represented below:

$$\begin{aligned} test\_sco_i = & \beta_0 + \beta_1 ICT\_Regulation_s + \beta_2 ICT\_Distraction_i + \beta_3 ICT\_Leisure_i \\ & + \beta_4 ESCS_i + \beta_5 low\_ses_i + \beta_6 inter\_reg\_lowses_i \\ & + \beta_7 inter\_dist\_lowses_i + \beta_8 female_i + \beta_9 private\_school_s \\ & + \sum \delta_c Country_c + u_i \end{aligned}$$

Lastly, we note that the Weighted Least Squares approach, instead of an ordinary one, aims to ensure maximum representativeness of our observations, not to cancel out heteroskedasticity, as the individual weights were provided in the database.

## 4 Violations of Assumptions

In this section, we evaluate the six Gauss-Markov assumptions underlying our models by checking for potential violations and implementing technical solutions.

### 4.1 Linearity in Parameters (MLR.1)

This assumption requires that all the parameters estimated in each of our various models are linear. Since, in our models, there is not a single  $\beta$  coefficient that appears multiplied, divided or as an exponent, we can conclude that this assumption holds.

### 4.2 Random Sampling (MLR.2)

PISA sample selection occurs in a two-stage process where it samples schools first and then students. To ensure that the report collects sufficient data about certain subsets of the student population that may not be so accessible, these groups may appear overrepresented in number of observations, which creates a clear violation to Random Sampling properties in Ordinary Least Squares regressions. Moreover, given that each observation comes with a specific weight that is determined by its estimated relative representativeness of the entire population, we decided to pursue a Weighted Least Squares approach. Therefore, we can claim that our results are representative of the broader student population in each of the selected countries.

### 4.3 No Perfect Collinearity (MLR.3)

To test if this assumption holds, we started by building the Correlation Matrix between all independent variables. As one can observe in Table 3, some correlations between interaction variables and their individual components reached high absolute values, as 0.88 and 0.79. However, we cannot simply eliminate these variables, as *inter\_reg\_lowses*, for example, is central to our research question, and we need to acknowledge that linear dependencies involving multiple variables may not be visible in the Correlation Matrix. In response, we conducted the Variance Inflation Test in the unweighted model 4 (see Table 9) and obtained individual scores below 5, except for *inter\_reg\_lowses* and *lowses* that corresponded to 6.99 and 8.57, indicating moderate multicollinearity. Since all the coefficients were successfully estimated for every model, we can reasonably conclude that the No Perfect Collinearity assumption is satisfied.

### 4.4 Zero Conditional Mean (MLR.4)

To tackle the Omitted Variable Bias that pushes the expected value of the error term to be different from 0, we gradually introduced sets of controls in our models that

worked as proxies for unobserved factors like home learning environments and student engagement, addressed national heterogeneity in our final regression, and assumed that school-level ICT Regulations were not the consequence of preceding Mathematics scores.

However, it is extremely difficult to eliminate all sources of bias: Some factors, like innate ability, cannot be directly measured. Others were simply not present in the database that we used to create our sample. Therefore, even though we tried to mitigate the OVB risk as much as possible, this assumption is unlikely to hold in our models, leading to biased and inconsistent estimates.

#### 4.5 Homoskedasticity (MLR.5)

Heteroskedasticity occurs when the variance of the error term is not constant across all possible levels of the independent variables ( $Var(u|X) \neq \sigma^2$ ). To check for it, we performed the Breusch-Pagan test (see Table 8) and identified that its Adjusted  $R^2$  of 3.47% indicates that some variance in squared residuals can be explained by our independent variables. In line with that, the resulting p-value approximates 0, leading us to reject the null hypothesis and confidently confirm the presence of heteroskedasticity.

To ensure that the significance levels reported in the next section are valid in the presence of heteroskedasticity, we relied on Robust Standard Errors and, as previously discussed, followed the Weighted Least Squares approach that helps address unequal variance by giving different weights to observations (associated with their sampling probability).

#### 4.6 Normality in Error Term (MLR.6)

Finally, the Normality in Error Term assumption states that the population error is independent of the explanatory variables and displays a normal distribution with mean of zero and constant variance of  $\sigma^2$ . To assess if this assumption holds in our final regression, we used a Normal Quantile-Quantile plot (see Figure 1) to compare the distribution of our residuals with a theoretical normal one. Although in the centre of the distribution our observations' estimated residuals follow the normal line, as they move further away from the centre, we have students whose scores were much different than those predicted by the model, resulting in increasingly steeper deviations from the line and higher-than-expected residuals. Despite that inconsistency with normality, we can invoke the Central Limit Theorem and claim that the distribution of the Weighted Least Squares estimators is asymptotically normal, given our large sample size of 81,850 observations. Therefore, it is not because of MLR.6 that our t-statistics and p-values lose their validity.

## 5 Results

In this section, we analyse the results of the multiple linear regressions conducted in Python. First, by studying the evolution of the various control models, we differentiate variables that are only significant because we are not accounting for the broader context from those that may have an actual effect on test scores. Then, we focus solely on the statistical interpretation of the main interest variable and finalize the section with a residual analysis.

### 5.1 Main Regression and Model Comparison

As Tables 5 and 7 show, the addition of controls between model 2 and 4 contributed to a 9.87 percentage point increase in the *Adjusted R*<sup>2</sup> to 24.07%. The biggest jump comes after including country fixed effects, highlighting its explanatory power. Furthermore, all our models are globally significant, with extremely low p-values. In particular, model 4 has an F-statistic of 1,442, which translates to a p-value that approximates 0. Its regression points to a Residual Standard Error of 742.7, which is much larger than the 81.7 obtained through OLS and does not seem reasonable given that 99% of all test scores fall within the 200 - 800 range. However, it happened because our student weights are not normalized, leading to an artificial increase equal to the scale of the weights.

Following the structure introduced in 3.3 to enumerate and describe control variables, we first comment on the demographic ones. In every model, we can observe that the socioeconomic index *ESCS* is one of the most powerful predictors. Regardless of the controls we added to our models, one standard-deviation increase in a student's socioeconomic condition yields the equivalent to nearly another two years of schooling — a 40-point jump in Mathematics scores — holding other factors constant. The *female* dummy showcased a statistically significant coefficient of -12.86 that did not react to the introduction of fixed country effects.

Then, regarding controls for Personal ICT habits, *ICT\_Distraction* reflects a significant (and slightly more negative across the models) score penalty for higher levels of distraction, reaching an average 5.57 decrease in test scores for each additional level of distraction in model 4, *ceteris paribus*. On the other hand, its interaction with low socioeconomic status only became statistically significant after accounting for country fixed effects and revealed a counterintuitive association: For this specific group, a one-unit increase in distraction translates on average to a 3.96 decrease in scores, 1.63 points more positive than for everyone else, holding other factors fixed, and challenging Bohnert

and Gracia's Differential Vulnerability hypothesis. Lastly, *ICT\_Leisure* presents a sign switch in model 4. Despite being significant and representing a consistent negative association across the first three models, once we included country fixed factors, its coefficient flipped from -3.96 to 3.37. This suggests that after considering negative cultural or national trends, *ICT\_Leisure* may act as a proxy for higher digital literacy.

Commenting now on our institutional control, the dummy *private\_school* was only introduced in model 3 and immediately stood out as the second most significant regressor in our model with a coefficient of 20.37 points. However, as we transitioned to model 4, Lubienski and Lubienski theory seemed to hold true: Not only did it switch sign to become -1.90 but also lost its significance for the 1% significance level (even though it remained significant for the 5% level).

To finalize this section, we have countries' dummy variables. As one can observe in Table 7, there is a clear difference in geographic performance between East Asian powerhouses, such as Macau and Hong Kong whose coefficients are 70.26 and 67.09, respectively, and Western nations where, for instance, the United States and Spain showcase a *ceteris paribus* association with test scores of -37.32 and -17.59. Other than Great Britain, our selected countries are significant for all relevant significance levels, suggesting that the country where these students live often works as a better score predictor than ICT personal habits or demographic differences.

## 5.2 The Interest Variables

Before moving to the interaction *inter\_reg\_lowses*, it is important to understand individual meaning of *ICT\_Regulation* in our models. From model 1 to 3, this variable was not particularly significant for our standard significance levels and oscillated in terms of sign. Then, in model 4, it became highly significant and displayed a coefficient of 5.08, indicating that a student that moves from a school with no regulation ( $x_{ICT\_Regulation} = 1$ ) to one that is heavily regulated in terms of ICT exposure and digital usage ( $x_{ICT\_Regulation} = 6$ ) experiences a predicted gain of 25.4 points, more than one additional schooling year. Hence, it is reasonable to conclude that regulation-related policies can have a tangible positive effect in Mathematics score outcomes.

Secondly, *low\_ses* 4.47 coefficient can be interpreted as the correction for the score prediction implicit in the *ESCS* slope. While in model 2 and 3, *ESCS* underpredicts the fall in scores for this subset of students, translated into a *low\_ses* coefficient of -8.55 in model 3 (only significant for the 10% significance level), in model 4 it overpredicts this fall in scores by 4.47 points for an increased significance level.



When it comes to the *ICT\_Regulation* and *low\_ses* interaction, not only do we have a significant association with scores across all models in which it is part of, but it also is consistently positive, although evidencing a decreasing trend. In model 4, this interaction coefficient of 1.84 reflects the average premium gain in scores per level of regulation that students from a disadvantaged social background obtain in comparison to the rest of the student pool, assuming all other factors are fixed. In their case, switching from a deregulated school to a heavily regulated one does not mean a predicted 25.4-point increase, but rather a total predicted boost of 34.6 points — an approximate half-year premium for this social group associated with heavy regulation.

### 5.3 Residual Analysis

To complement and add some narrative to the residual analysis conducted in 4.6 Normality in Error Term, we generated a Residuals Histogram (see Figure 2). Its visualization confirms that standardized residuals centre almost perfectly around 0, meaning that, on average, our final model does not systematically over-predict or under-predict Mathematics scores — it is unbiased in its central tendency. We can also observe a sharper peak and heavier tails than the ones associated with the normal distribution, evidencing Leptokurtosis (high concentration around the centre), which, in turn, signals that for most of our observed students, the model is very accurate and explains the typical student reality extremely well. However, it is worth noting that standardized residuals reaching absolute values of almost five points to the existence of extreme outliers, reflecting the complexity of correctly estimating individual performance. Hence, those few students that our model completely failed in predicting test scores are the ones pushing down a potentially more substantial Adjusted  $R^2$ .

Finally, our residual analysis can only be complete with the introduction of the Studentized-residuals-versus-fitted plot (check Figure 3). The LOWESS line represents the local mean of studentized residuals relative to each predicted test score. As it remains flat and close to zero for most of the distribution's density area and the residuals' cloud does not exhibit any obvious quadratic shape (or other type of systematic curvature), we can say that the Weighted-Least-Squares linear form was adequate for most of the sample. Nonetheless, on both ends, the line drifts away from 0, showcasing that low-score achievers had their scores under-predicted, while those of high-achievers were over-predicted. Thus, reinforcing the idea that this regression works better for the “typical” student than for exceptionally good and bad performers.



## 6 Conclusion

We started this project with an intent to analyse the relationship between digital school-level regulation and the Mathematics test score differential between socioeconomically disadvantaged students and their more affluent peers, and now conclude with some interesting statistically significant results, policy recommendations and the acknowledgement of the most relevant limitations of our study.

First, ICT Regulation is positively associated with test scores. The transition from a low regulatory environment to a very high one is, on average and *ceteris paribus*, equivalent to more than one additional schooling year in terms of test performance. Although, the relationship only became statistically significant after accounting for country fixed effects. More interestingly, our results support our initial thesis that not only is regulation expected to increase scores, but it also contributes to a narrower performance gap between socioeconomically disadvantaged students and their more affluent peers — its interaction term with lower-SES background remained consistently positive and significant across the three models it was part of.

Second, we have found both points of convergence and divergence with the current academic literature. On one hand, we could confirm that socioeconomic background is a very powerful predictor of performance, only behind the country in which the student's test is observed. On the other hand, we challenge the accuracy of the Differential Vulnerability Hypothesis in the sense that we did not find sufficient evidence to support the claim that disadvantaged students suffer more from ICT Distraction than the rest of the student pool. In fact, their interaction term points to a significant positive association in our final regression.

Third, our study clearly supports further and stricter legislation on digital access and usage at the school for all the potential benefits one can imagine: from more meaningful and real social connections to an increased capacity of focusing solely on the contents taught in class for longer periods. However, our recommendations must be taken with a grain of salt. Our model showcased extremely competent predictive accuracy for the median fifteen-year-old student. But it also evidenced that for remarkably bad and good performers, our linear Weighted-Least-Squares approach may not be the best to draft confident conclusions, with studentized residuals drifting away from zero on the extremes. Not to mention that the spectre of the Omitted Variable Bias (and implicit violation of MLR.4) haunts most empirical studies conducted in social sciences. In our case, it is no different.

## 7 Bibliography

Ji, Zhiqiang, and Jia Pang. 2023. "Social Media and Student Academic Performance: A Cross-Country Analysis Using PISA 2018." *International Journal of Educational Technology and Learning*, 15 (1).

Navarro-Martinez, Oscar, and Beatriz Peña-Acuña. 2022. "Technology Usage and Academic Performance in the PISA 2018 Report." *Journal of New Approaches in Educational Research*, 11 (1).

Vizmonte, Emily Jo, and Dominic Vincent Ligot. 2024. "When Digital Trace Becomes Educational Waste: Is There a Trade-Off Between Time Spent on the Internet and Social Media and Educational Performance?" *SSRN Electronic Journal*.

Maghnouj, Soumaya, et al. 2025. "Finite Time to Learn and Play: Whole Student Development and Students' Digital Leisure Outside of School." *OECD Education Policy Perspectives*, No. 130. Paris: OECD Publishing.

Bohnert, Martin, and Pablo Gracia. 2024. "Digital Distraction, Attention Regulation, and Inequality." *Philosophy & Technology*, 37.

Lubienski, Sarah Theule, and Christopher Lubienski. 2006. "School Sector and Academic Achievement: A Multilevel Analysis of NAEP Mathematics Data." *American Educational Research Journal*, 43 (4).

Hargittai, Eszter. 2008. "The Digital Reproduction of Inequality." In *Social Stratification*, edited by David Grusky. Boulder: Westview Press.

Sirin, Selcuk R. 2005. "Socioeconomic Status and Academic Achievement: A Meta-Analytic Review of Research." *Review of Educational Research*, 75 (3): 417–453

## 8 Appendix

|                   | Variable                                                     |
|-------------------|--------------------------------------------------------------|
| test_sco          | PISA 2022 Mathematics test score                             |
| ESCS              | Socioeconomic continuous index                               |
| low_ses           | Bottom 25% observations from ESCS                            |
| female            | The observed student is a female                             |
| private_school    | The observed student studies at a private school             |
| ICT_Regulation    | Scale of school-level ICT regulation                         |
| ICT_Leisure       | Scale of ICT usage for fun purposes outside of the classroom |
| ICT_Distracton    | Scale of ICT distraction during classes                      |
| Student_Weight    | Representativeness weight attributed to each observation     |
| inter_reg_lowses  | Interaction variable between ICT_Regulation and low_ses      |
| inter_dist_lowses | Interaction variable between ICT_Distracton and low_ses      |
| Country_ESP       | The observed student studies in Spain                        |
| Country_GBR       | The observed student studies in Great Britain                |
| Country_HKG       | The observed student studies in Hong Kong                    |
| Country_ITA       | The observed student studies in Italy                        |
| Country_JPN       | The observed student studies in Japan                        |
| Country_KOR       | The observed student studies in Korea                        |
| Country_MAC       | The observed student studies in Macao                        |
| Country_TAP       | The observed student studies in Taiwan                       |
| Country_USA       | The observed student studies in the United States            |

*Table 1. Variables*

|                 | test_sco | ICT_Regulation | ICT_Distracton | ICT_Leisure |
|-----------------|----------|----------------|----------------|-------------|
| Mean            | 492.197  | 3.527          | 2.770          | 3.594       |
| Median          | 491.471  | 4.000          | 3.000          | 4.000       |
| Std. Deviation  | 94.766   | 1.419          | 1.731          | 1.489       |
| Sample Variance | 8980.514 | 2.014          | 2.995          | 2.218       |
| Kurtosis        | -0.395   | -0.870         | -1.301         | -1.000      |
| Skewness        | 0.081    | -0.482         | 0.382          | -0.452      |
| Range           | 683.258  | 5.000          | 5.000          | 5.000       |
| Minimum         | 163.673  | 1.000          | 1.000          | 1.000       |
| Maximum         | 846.932  | 6.000          | 6.000          | 6.000       |
| Count           | 81850    | 81850          | 81850          | 81850       |

*Table 2a. Descriptive statistics*

|                 | ESCS   | low_ses | inter_reg_lowses | inter_dist_lowses |
|-----------------|--------|---------|------------------|-------------------|
| Mean            | 0.043  | 0.221   | 0.712            | 0.583             |
| Median          | 0.134  | 0.000   | 0.000            | 0.000             |
| Std. Deviation  | 0.929  | 0.415   | 1.519            | 1.375             |
| Sample Variance | 0.863  | 0.172   | 2.308            | 1.890             |
| Kurtosis        | 0.077  | -0.186  | 2.577            | 5.748             |
| Skewness        | -0.482 | 1.347   | 2.000            | 2.587             |
| Range           | 13.776 | 1.000   | 6.000            | 6.000             |
| Minimum         | -6.396 | 0.000   | 0.000            | 0.000             |
| Maximum         | 7.380  | 1.000   | 6.000            | 6.000             |
| Count           | 81850  | 81850   | 81850            | 81850             |

*Table 2b. Descriptive statistics*

|                 | female | private_school |
|-----------------|--------|----------------|
| Mean            | 0.504  | 0.176          |
| Median          | 1.000  | 0.000          |
| Std. Deviation  | 0.500  | 0.381          |
| Sample Variance | 0.250  | 0.145          |
| Kurtosis        | -2.000 | 0.904          |
| Skewness        | -0.016 | 1.704          |
| Range           | 1.000  | 1.000          |
| Minimum         | 0.000  | 0.000          |
| Maximum         | 1.000  | 1.000          |
| Count           | 81850  | 81850          |

*Table 2c. Descriptive statistics*

|                   | test_sco | ICT_Regulation | ICT_Distracton | ICT_Leisure |
|-------------------|----------|----------------|----------------|-------------|
| test_sco          | 1.000    |                |                |             |
| ICT_Regulation    | 0.028    | 1.000          |                |             |
| ICT_Distracton    | -0.066   | 0.218          | 1.000          |             |
| ICT_Leisure       | -0.042   | 0.431          | 0.061          | 1.000       |
| ESCS              | 0.360    | 0.172          | 0.047          | 0.088       |
| low_ses           | -0.262   | -0.112         | -0.040         | -0.045      |
| inter_reg_lowses  | -0.229   | 0.140          | 0.032          | 0.050       |
| inter_dist_lowses | -0.228   | 0.001          | 0.259          | 0.002       |
| female            | -0.077   | -0.045         | 0.022          | 0.010       |
| private_school    | 0.108    | -0.081         | 0.073          | -0.149      |

*Table 3a. Correlation Matrix*

|                   | ESCS   | low_ses | inter_reg_lowses | inter_dist_lowses |
|-------------------|--------|---------|------------------|-------------------|
| ESCS              | 1.000  |         |                  |                   |
| low_ses           | -0.760 | 1.000   |                  |                   |
| inter_reg_lowses  | -0.673 | 0.881   | 1.000            |                   |
| inter_dist_lowses | -0.608 | 0.796   | 0.786            | 1.000             |
| female            | -0.003 | 0.012   | 0.003            | 0.011             |
| private_school    | 0.066  | -0.055  | -0.058           | -0.037            |

*Table 3b. Correlation Matrix*

|                | female | private_school |
|----------------|--------|----------------|
| female         | 1.000  |                |
| private_school | 0.002  | 1.000          |

*Table 3c. Correlation Matrix*

| Variable            | Estimate | Robust std. Error | t value | Pr(> t ) |
|---------------------|----------|-------------------|---------|----------|
| (Intercept)         | 516.883  | 2.260             | 228.708 | < 2e-16  |
| ICT_Regulation      | 1.032    | 0.548             | 1.884   | 0.060    |
| ICT_Distracton      | -4.503   | 0.448             | -10.060 | < 2e-16  |
| ICT_Leisure         | -4.859   | 0.498             | -9.759  | < 2e-16  |
| ESCS                | 37.485   | 0.790             | 47.427  | < 2e-16  |
| Residual std. Error |          |                   | 789.792 |          |
| Adj. R-Squared      |          |                   | 0.141   |          |
| F-statistic         |          |                   | 591.937 |          |

*Table 4. Model 1 regression*

| Variable            | Estimate | Robust std. Error | t value | Pr(> t )  |
|---------------------|----------|-------------------|---------|-----------|
| (Intercept)         | 519.877  | 2.622             | 198.266 | < 2.2e-16 |
| ICT_Regulation      | -0.148   | 0.624             | -0.237  | 0.813     |
| ICT_Distracton      | -4.727   | 0.503             | -9.392  | < 2.2e-16 |
| ICT_Leisure         | -4.836   | 0.496             | -9.753  | < 2.2e-16 |
| ESCS                | 40.275   | 1.224             | 32.908  | < 2.2e-16 |
| low_ses             | -8.551   | 4.539             | -1.884  | 0.06      |
| inter_reg_lowses    | 4.223    | 1.215             | 3.475   | 0.001     |
| inter_dist_lowses   | 0.803    | 1.098             | 0.731   | 0.465     |
| Residual std. Error |          |                   | 789.222 |           |
| Adj. R-Squared      |          |                   | 0.143   |           |
| F-statistic         |          |                   | 349.31  |           |

*Table 5. Model 2 regression*

| Variable            | Estimate | Robust std. Error | t value | Pr(> t ) |
|---------------------|----------|-------------------|---------|----------|
| (Intercept)         | 520.818  | 2.818             | 184.8   | < 0.001  |
| ICT_Regulation      | -0.121   | 0.621             | -0.194  | 0.846    |
| ICT_Distracton      | -5.028   | 0.503             | -10.005 | < 0.001  |
| ICT_Leisure         | -3.955   | 0.493             | -8.02   | < 0.001  |
| ESCS                | 39.742   | 1.216             | 32.678  | < 0.001  |
| low_ses             | -7.78    | 4.512             | -1.724  | 0.085    |
| inter_reg_lowses    | 3.951    | 1.207             | 3.273   | 0.001    |
| inter_dist_lowses   | 1        | 1.093             | 0.914   | 0.36     |
| female              | -13.914  | 1.468             | -9.477  | < 0.001  |
| private_school      | 20.365   | 1.612             | 12.636  | < 0.001  |
| Residual std. Error |          |                   | 783.777 |          |
| Adj. R-Squared      |          |                   | 0.154   |          |
| F-statistic         |          |                   | 313.218 |          |

*Table 6. Model 3 regression*

| Variable               | Estimate | Robust std. Error | t value | Pr(> t ) |
|------------------------|----------|-------------------|---------|----------|
| (Intercept)            | 488.584  | 1.511             | 323.256 | < 2e-16  |
| ICT_Regulation         | 5.084    | 0.279             | 18.245  | < 2e-16  |
| ICT_Distracton         | -5.578   | 0.196             | -28.455 | < 2e-16  |
| ICT_Leisure            | 3.366    | 0.249             | 13.498  | < 2e-16  |
| ESCS                   | 41.621   | 0.487             | 85.499  | < 2e-16  |
| low_ses                | 4.467    | 1.96              | 2.279   | 0.023    |
| inter_reg_lowses       | 1.842    | 0.489             | 3.771   | 0.000    |
| inter_dist_lowses      | 1.622    | 0.412             | 3.936   | 0.000    |
| female                 | -12.86   | 0.579             | -22.219 | < 2e-16  |
| private_school         | -1.896   | 0.88              | -2.155  | 0.031    |
| Country_CodeESP        | -17.591  | 1.596             | -11.024 | < 2e-16  |
| Country_CodeGBR        | -2.205   | 1.569             | -1.405  | 0.160    |
| Country_CodeHKG        | 67.089   | 3.763             | 17.829  | < 2e-16  |
| Country_CodeITA        | -15.281  | 1.517             | -10.072 | < 2e-16  |
| Country_CodeJPN        | 47.562   | 1.341             | 35.472  | < 2e-16  |
| Country_CodeKOR        | 26.213   | 1.584             | 16.551  | < 2e-16  |
| Country_CodeMAC        | 70.256   | 11.281            | 6.228   | 0.000    |
| Country_CodeTAP        | 59.196   | 2.043             | 28.979  | < 2e-16  |
| Country_CodeUSA        | -37.325  | 1.209             | -30.882 | < 2e-16  |
| Residual std. Error    |          |                   | 742.7   |          |
| Adj. R-Squared         |          |                   | 0.241   |          |
| F-statistic (81831 DF) |          |                   | 542.669 |          |

*Table 7. Model 4 regression*

| Test Name           | Studentized Breusch-Pagan |
|---------------------|---------------------------|
| LM Stat             | 2855.457                  |
| LM p-value          | 0                         |
| F-statistic         | 164.332686                |
| F-statistic p-value | 0                         |
| Adj. R-Squared      | 0.0347                    |

*Table 8. Model 4 Unweighted Studentized Breusch-Pagan test*



| Variable          | Variance Inflation Factor |
|-------------------|---------------------------|
| Intercept         | 31.950                    |
| ICT_Regulation    | 1.771                     |
| ICT_Distracton    | 1.460                     |
| ICT_Leisure       | 1.219                     |
| ESCS              | 2.578                     |
| low_ses           | 8.574                     |
| inter_reg_lowses  | 6.993                     |
| inter_dist_lowses | 4.267                     |
| female            | 1.011                     |
| private_school    | 1.177                     |
| Country_ESP       | 4.711                     |
| Country_GBR       | 2.768                     |
| Country_HKG       | 2.168                     |
| Country_ITA       | 2.673                     |
| Country_JPN       | 2.120                     |
| Country_KOR       | 2.160                     |
| Country_MAC       | 1.838                     |
| Country_TAP       | 2.085                     |
| Country_USA       | 1.835                     |

*Table 9. Model 4 Unweighted Variance Inflation Factor*

| Variable               | Estimate | Robust std. Error | t value | Pr(> t ) |
|------------------------|----------|-------------------|---------|----------|
| (Intercept)            | 493.050  | 1.614             | 305.455 | < 2e-16  |
| ICT_Regulation         | 6.724    | 0.277             | 24.272  | < 2e-16  |
| ICT_Distracton         | -4.279   | 0.204             | -20.962 | < 2e-16  |
| ICT_Leisure            | -1.711   | 0.217             | -7.884  | 3.21e-15 |
| ESCS                   | 37.335   | 0.488             | 76.439  | < 2e-16  |
| low_ses                | -1.336   | 1.948             | -0.686  | 0.493    |
| inter_reg_lowses       | 1.834    | 0.483             | 3.799   | 0.001    |
| inter_dist_lowses      | 2.212    | 0.406             | 5.454   | 4.94e-08 |
| female                 | -11.860  | 0.574             | -20.657 | < 2e-16  |
| private_school         | 2.771    | 0.693             | 4.000   | 6.35e-05 |
| Country_CodeESP        | -10.427  | 1.315             | -7.930  | 2.21e-15 |
| Country_CodeGBR        | -8.633   | 1.472             | -5.866  | 4.49e-09 |
| Country_CodeHKG        | 68.609   | 1.707             | 40.194  | < 2e-16  |
| Country_CodeITA        | -13.305  | 1.460             | -9.112  | < 2e-16  |
| Country_CodeJPN        | 49.019   | 1.655             | 29.618  | < 2e-16  |
| Country_CodeKOR        | 29.334   | 1.603             | 18.297  | < 2e-16  |
| Country_CodeMAC        | 71.224   | 1.729             | 41.191  | < 2e-16  |
| Country_CodeTAP        | 49.950   | 1.636             | 30.542  | < 2e-16  |
| Country_CodeUSA        | -29.749  | 1.822             | -16.330 | < 2e-16  |
| Residual std. Error    |          | 81.7              |         |          |
| Adj. R-Squared         |          | 0.218             |         |          |
| F-statistic (81831 DF) |          | 1190.933          |         |          |

*Table 10. Model 4 regression using Ordinary Least Squares*

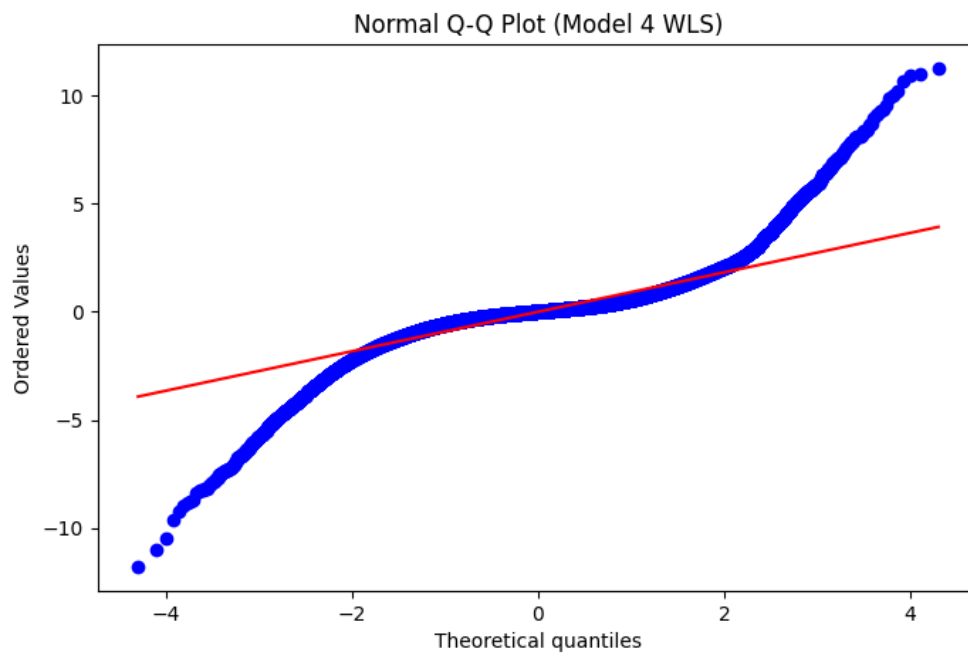


Figure 1. Model 4 normal Q-Q plot

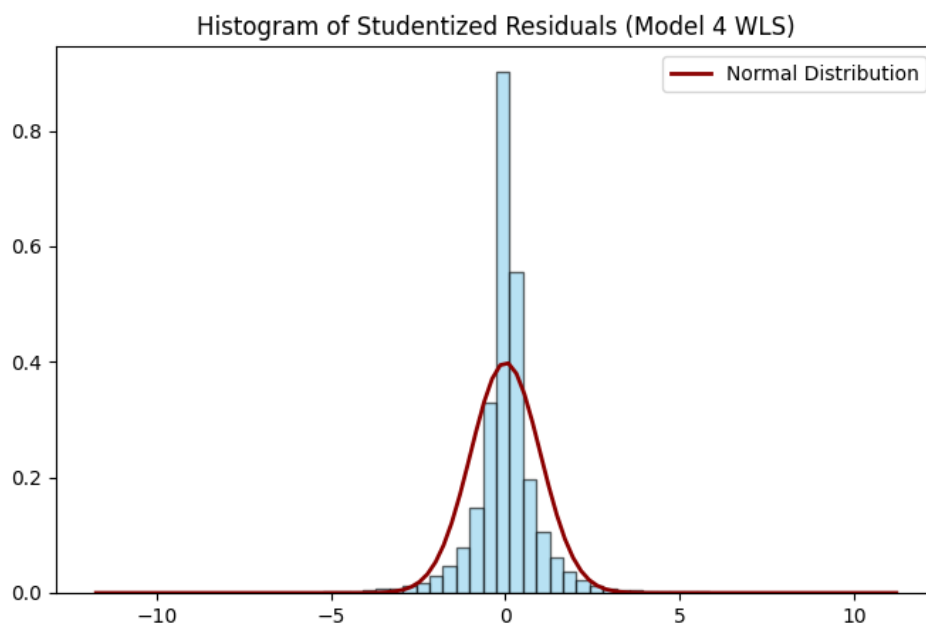
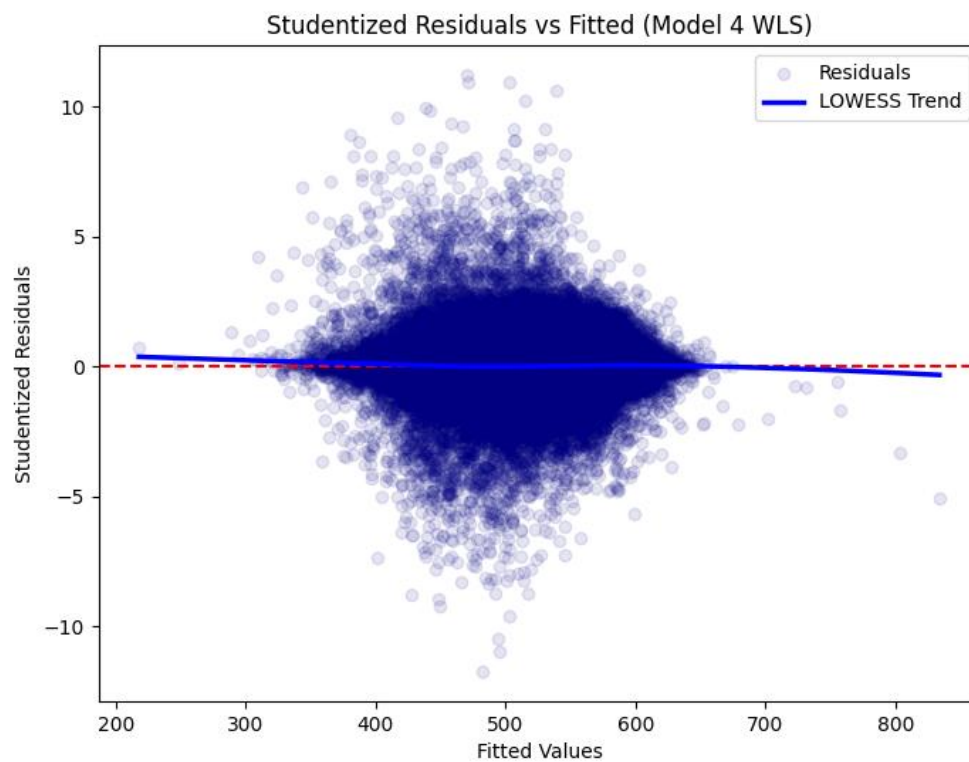


Figure 2. Histogram of residuals



*Figure 3. Studentized-residuals-versus-fitted plot*